

Delta-one basis and lead-lag: ETF, index futures and the cash basket - fair value with measured financing and dividends, mean reversion, who leads whom, and what transfers from ES/SPY to a new product

Question. A delta-one desk carries an index as a future, a fund and a basket and lives off the differences between them. This project measures, on free data, where the basis sits against a fair value whose financing (the bill curve, EFR/SOFR) and dividends (a point-in-time forecast with a within-quarter accrual profile read off the fund's own price) are measured rather than assumed; how fast a deviation closes at daily and minute horizons; the lead-lag between future, fund, basket and cash index by the Hayashi-Yoshida / Hoffmann-Rosenbaum-Vetter estimator on 23 days of 1-minute bars and one year of ETF-versus-index minutes; a basis strategy net of costs; and the new-products test, in which everything fitted on ES/SPY is applied unchanged to NQ/QQQ, RTY/IWM and a Euro Stoxx 50 pair.

Answer. (1) Half-way through a dividend quarter 42% of SPY's dividends have gone ex; with that profile the SPY premium to fair has sd 16.8 bp against 23.9 with no accrual model; the dividend forecast misses by 6% on SPY and 32% on IWM, so the model helps the first and hurts the second. (2) With both legs marked at 16:00 (505 days from 2024-04-29) the ES front implies financing of bills + 50 bp (NQ +42, RTY +71); the calendar spread implies bills + 67. (3) Richness has an OLS half-life of 1.2 days on ES and a noise-robust persistent component of 25 days; within the day the minute deviation halves in 1.4 min [0.9, 2.3]. (4) At one-minute resolution future and fund are simultaneous (LLR 0.97 [0.94, 1.00]); the fund leads the cash index by the grid's first step with LLR 1.12 on SPY/SPX, 1.24 on QQQ/NDX and 3.96 [3.50, 4.31] on IWM/RUT: the index's stale-price lag grows with the number of small names. (5) The z-score basis strategy with ES-fitted parameters nets 34 bp/yr [2, 74] on ES, 62 [20, 107] on NQ, 77 [15, 171] on RTY; the same rule on a series whose legs are marked an hour apart 'earns' 327 bp/yr, an artefact shown beside the clean number. (6) Transfer: NQ: profile transfers; dynamics transfer; strategy transfers; lead-lag direction transfers, size does not. RTY: accrual model hurts (no model is best); dynamics transfer; strategy transfers; lead-lag direction transfers, size does not. SX5E: profile transfers; lead-lag does not transfer.

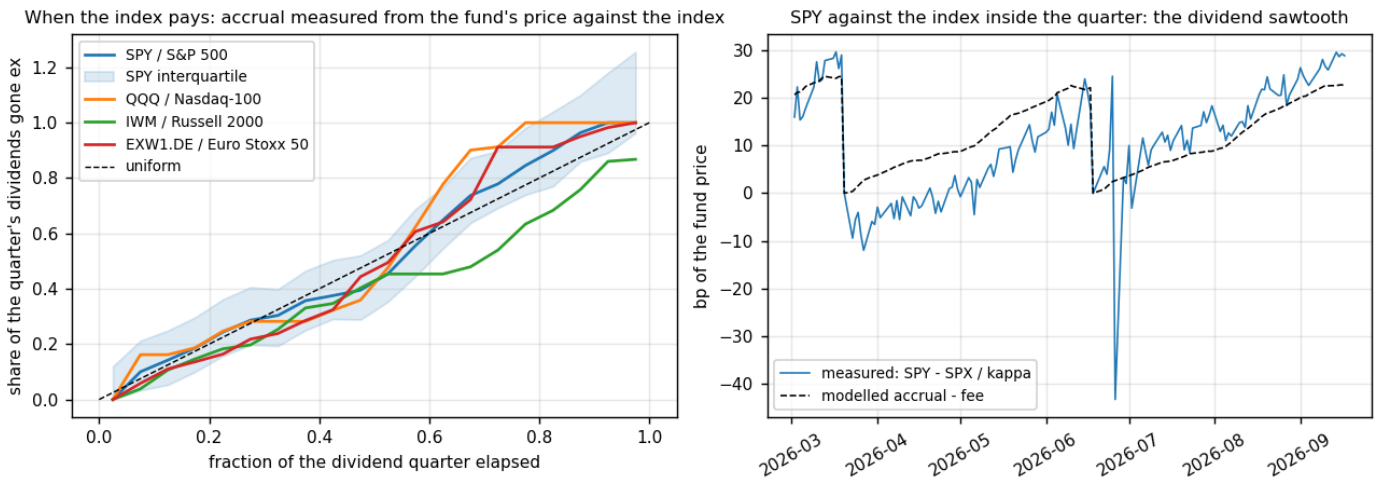


Figure 1. Left: cumulative share of the quarter's dividends gone ex by fraction of the quarter, measured from each fund's price against its index. Right: SPY against the index inside the last two quarters, measured and modelled.

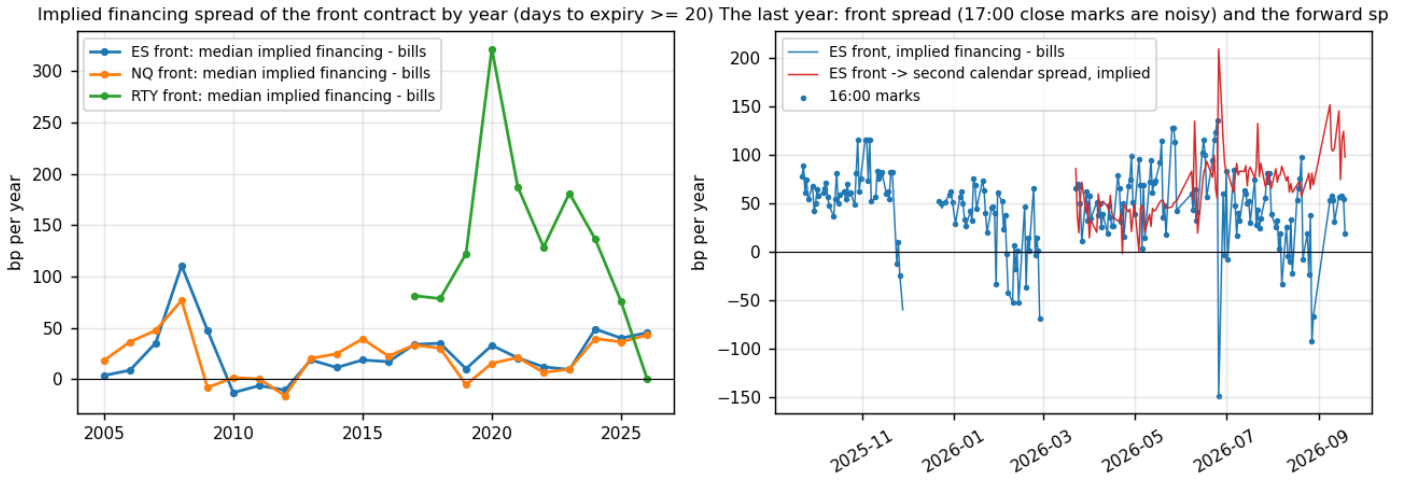


Figure 2. Left: median implied financing spread of the front contract over bills, by year, from the daily closes. Right: the last year of the ES front spread with the 16:00 marks, and the calendar-spread-implied forward financing.

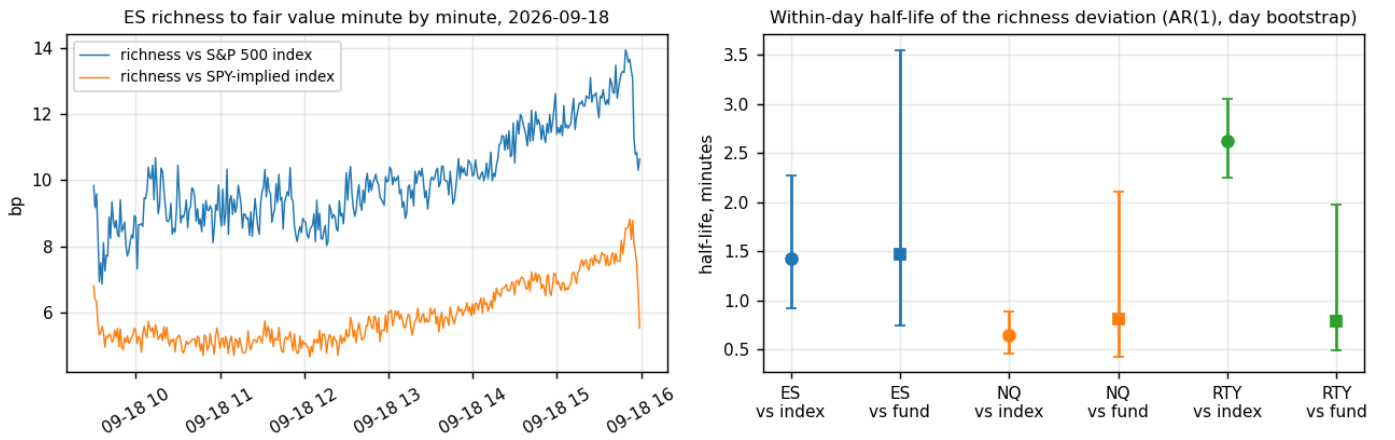


Figure 3. Left: ES richness to fair value minute by minute on the last archived day, against the index and the fund-implied index. Right: within-day half-life of the richness deviation by pair.

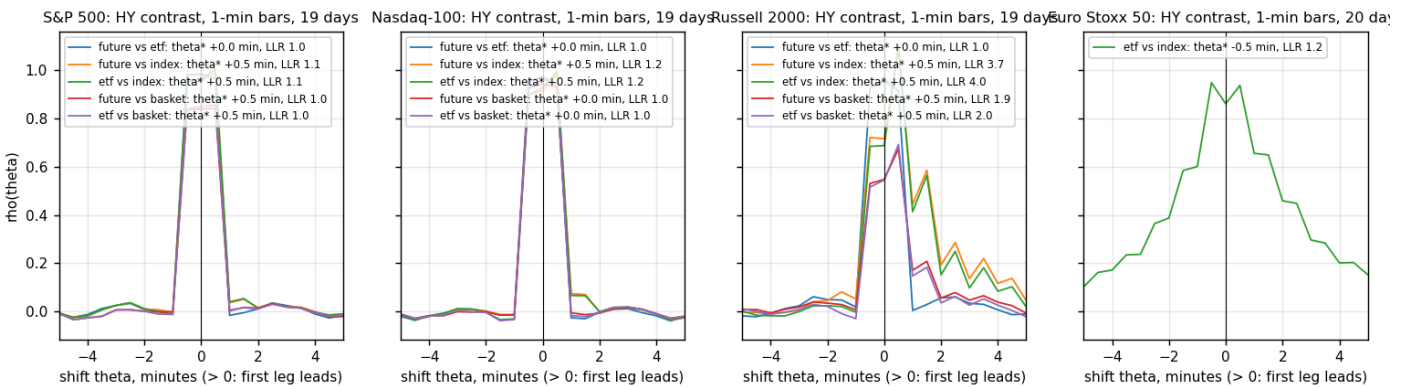


Figure 4. The Hayashi-Yoshida contrast $\rho(\theta)$ on 1-minute bars, per pair and leg pair.

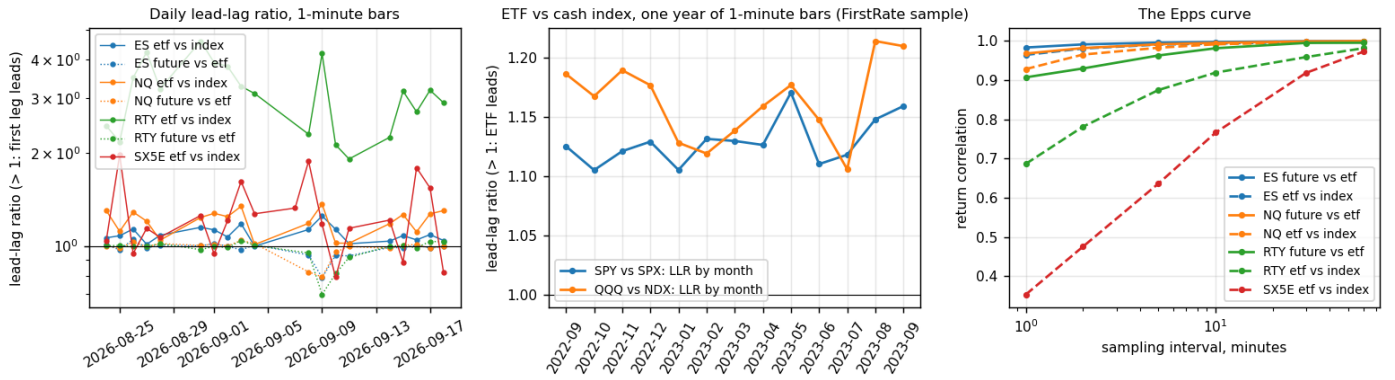


Figure 5. Left: the daily lead-lag ratio. Middle: ETF vs cash index by month over the FirstRate year. Right: the Epps curve.

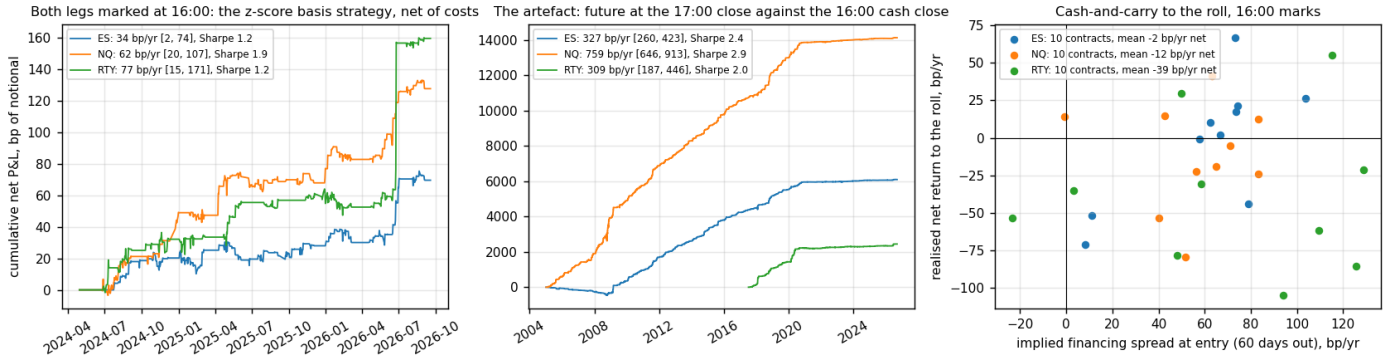


Figure 6. Left: the z-score basis strategy on the 16:00-marked series, net of costs. Middle: the marking artefact. Right: cash-and-carry to the roll.

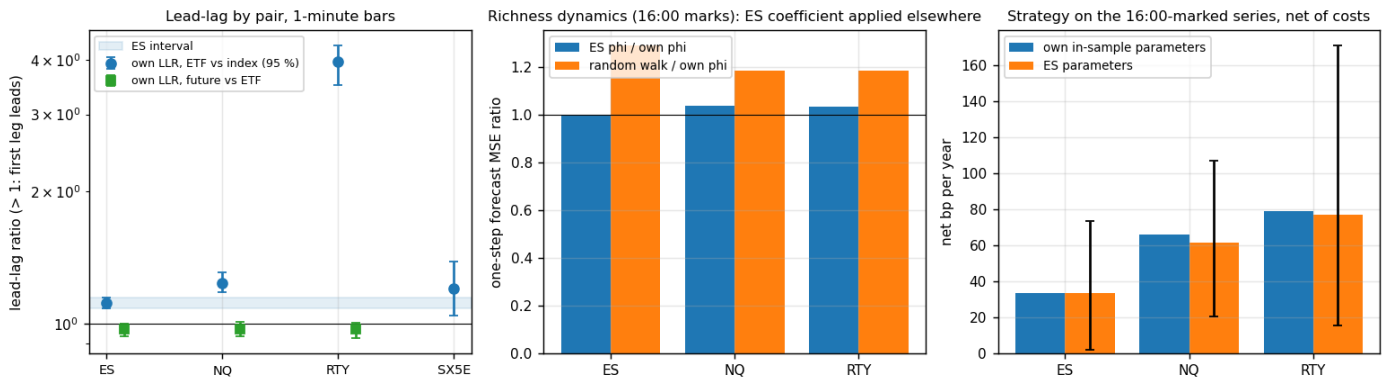


Figure 7. The transfer table: lead-lag ratio, richness dynamics and strategy return by pair, with ES's parameters applied to the others.

Fair value and mean reversion, 16:00 marks

pair	days	rho mean	rho sd	spread median	spread sd	fwd spread	HL OLS	HL lo	HL hi	HL robust
ES	505	6.95	7.21	49.82	51.00	66.70	1.17	0.66	1.65	24.62
NQ	505	6.16	7.11	41.70	47.68	65.79	1.87	1.16	2.19	15.24
RTY	505	9.83	10.33	70.71	105.84	5.35	1.85	1.11	2.25	8.63

Fund premium to the index

fund	days	sd all	sd 5y	SPY profile	uniform	no accrual	HL days	div MAE %
SPY	5409	16.81	6.92	16.81	17.02	23.91	1.77	6.17
QQQ	5284	11.65	5.05	11.62	11.72	12.59	0.93	20.96
IWM	5402	19.38	13.31	21.40	20.78	18.78	1.27	31.94
EXW1.DE	3377	51.97	48.78	51.93	51.80	53.80	16.05	75.25

Lead-lag, 1-minute bars (theta* > 0: first leg leads)

pair	legs	days	theta*	lo	hi	rho*	rho(0)	LLR	LLR lo	LLR hi
ES	future vs etf	19	0.00	-30.00	0.00	0.98	0.98	0.97	0.94	1.00
ES	future vs index	19	30.00	0.00	30.00	0.99	0.96	1.09	1.06	1.12
ES	etf vs index	19	30.00	30.00	30.00	1.00	0.96	1.12	1.08	1.15
ES	future vs basket	19	30.00	-30.00	30.00	0.84	0.84	1.02	0.97	1.05
ES	etf vs basket	19	30.00	0.00	30.00	0.86	0.85	1.04	1.01	1.07
ES	basket vs index	19	30.00	-30.00	30.00	0.88	0.87	1.02	0.99	1.06
ES	SPY vs SPX (1y)	251	30.00	30.00	30.00	1.03	0.97	1.13	1.12	1.14
NQ	future vs etf	19	0.00	-30.00	0.00	0.97	0.97	0.97	0.94	1.01
NQ	future vs index	19	30.00	30.00	30.00	0.98	0.92	1.21	1.15	1.29
NQ	etf vs index	19	30.00	30.00	30.00	0.99	0.93	1.24	1.18	1.31
NQ	future vs basket	19	0.00	-30.00	30.00	0.94	0.94	1.01	0.98	1.05
NQ	etf vs basket	19	0.00	0.00	30.00	0.96	0.96	1.03	1.01	1.06
NQ	basket vs index	19	30.00	30.00	30.00	0.98	0.93	1.17	1.12	1.23
NQ	QQQ vs NDX (1y)	251	30.00	30.00	30.00	1.06	0.97	1.18	1.17	1.19
RTY	future vs etf	19	0.00	-30.00	0.00	0.96	0.96	0.97	0.93	1.00
RTY	future vs index	19	30.00	30.00	30.00	1.09	0.72	3.74	3.18	4.12
RTY	etf vs index	19	30.00	30.00	30.00	1.10	0.69	3.96	3.50	4.31
RTY	future vs basket	19	30.00	30.00	30.00	0.67	0.55	1.90	1.60	2.14
RTY	etf vs basket	19	30.00	30.00	30.00	0.69	0.55	2.03	1.72	2.29
RTY	basket vs index	19	30.00	30.00	30.00	0.88	0.63	1.72	1.44	1.96
SX5E	etf vs index	20	-30.00	-30.00	30.00	0.95	0.86	1.20	1.04	1.39

Strategy, net of costs

pair	series	days	trades	trade net	hit	net bp/yr	lo	hi	Sharpe	max DD
ES	16:00 marks	522	64	1.09	0.62	33.54	2.10	73.54	1.21	14.88
ES	17:00 closes (artefact)	4695	586	10.41	0.73	327.39	259.84	423.03	2.42	466.26
NQ	16:00 marks	522	62	2.06	0.71	61.54	20.32	106.72	1.92	12.53
NQ	17:00 closes (artefact)	4695	629	22.47	0.90	758.71	646.44	912.90	2.91	385.39
RTY	16:00 marks	522	52	3.06	0.60	76.90	15.32	170.69	1.20	16.60
RTY	17:00 closes (artefact)	1997	219	11.17	0.70	308.67	187.21	446.08	2.01	66.76

Validation and limits

Fair value reproduces $F^* = I(1 + r \text{ tau}) - D$ on a hand case and the implied financing of the fair price is the input rate; the calendar reproduces the third Fridays including the Juneteenth pull-forward; AR(1) recovers a known phi and the autocovariance-ratio estimator recovers it under heavy measurement noise where OLS does not; the Hayashi-Yoshida sum equals the realised covariance on synchronous data and the HRV estimator recovers a known lag on Poisson-sampled Brownian motions; the strategy P&L reconciles to the basis move and the calendar-day carry with costs on entry, exit and roll; the Databento loader is tested on a synthetic file. Limits: 1-minute bars are the finest free data, so the seconds-scale future-vs-fund lead is invisible and the ETF-vs-index theta* is the grid's first step; the 16:00-marked series is two and a half

years long and the strategy's mask sensitivity is large; the baskets are the funds' largest names at fixed weights; the Euro Stoxx pair has no free future and a thinly traded fund; the dividend forecasts come from the funds' distributions and their errors are stated per fund. No Databento depth was bought.